QUASI-COMPACTNESS TRANSFERS BETWEEN A DILATATION AND ITS FORMAL MODEL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces etale locally to the affine case through the functoriality of the completion functor, extracts a finite subcover of the formal part from quasi-compactness of W, and covers X' by the open part plus the finitely many pieces mapping onto W_red

Editorial note. The label lemma-output-quasi-compact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the dilatations theorem (theorem-dilatations-general of the same chapter) and its correspondence between f : X' -> X and g : W -> X_{/T}
- quasi-compactness of f implies quasi-compactness of g (lemma-quasi-compact-gives-quasi-compact of the same chapter)
- quasi-compactness of morphisms of algebraic spaces checked etale locally on the base (Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-local)
- quasi-compactness of morphisms of formal algebraic spaces characterized etale locally (Formal Spaces, Lemma formal-spaces-lemma-characterize-quasi-compact-morphism) and the functoriality of the completion functor (lemma-functoriality-completion-functor of the same chapter)
- surjectivity of etale covers and the description |X'| = |U| amalgramated with |W_red| for the dilatation

Source reference. Original lemma 0AJB: sources/stacks-chow/tags.txt, source lines 13294–13294. Prerequisite pointer invoked at proof line(s) 7132; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDQ: sources/stacks-restricted/pinned-original-source.tex, source lines 6790–6790. Prerequisite pointer invoked at proof line(s) 7135; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDE: sources/stacks-restricted/pinned-original-source.tex, source lines 6067–6067. Prerequisite pointer invoked at proof line(s) 7125; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03KG: sources/stacks-chow/tags.txt, source lines 4676–4676. Prerequisite pointer invoked at proof line(s) 7130; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ARB: sources/stacks-restricted/pinned-original-source.tex, source lines 7015–7015. Prerequisite pointer invoked at proof line(s) 7118; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With assumptions and notation as in Theorem 0ARB let $f : X' \to X$ correspond to $g : W \to X_{/T}$. Then f is quasi-compact if and only if g is quasi-compact.*

Proof. If f is quasi-compact, then g is quasi-compact by Lemma 0GDE. Conversely, assume g is quasi-compact. Choose an étale covering $\{X_i \to X\}$ with X_i affine. It suffices to prove that the base change $X' \times_X X_i \to X_i$ is quasi-compact, see Morphisms of Spaces, Lemma 03KG. By Formal Spaces, Lemma 0AJB the base changes $W_i \times_{X_{/T}} (X_i)_{/T} \to (X_i)_{/T}$ are quasi-compact. By Lemma 0GDQ we reduce to the case described in the next paragraph.

Assume X is affine and $g : W \to X_{/T}$ quasi-compact. We have to show that X' is quasi-compact. Let $V \to X'$ be a surjective étale morphism where $V = \coprod_{j \in J} V_j$ is a disjoint union of affines. Then $V_{/T} \to X'_{/T} = W$ is a surjective étale morphism. Since W is quasi-compact, then we can find a finite subset $J' \subset J$ such that $\coprod_{j \in J'} (V_j)_{/T} \to W$ is surjective. Then it follows that

$$U \amalg \coprod\nolimits_{j \in J'} V_j \longrightarrow X'$$

is surjective (and hence X' is quasi-compact). Namely, we have $|X'| = |U| \amalg |W_{red}|$ as $X'_{/T} = W$. □